5.0 Introduction

It is well known that how important are the wheels and tyres in automobiles. The vehicle can be pulled even with no engine. However, without the wheels the vehicle cannot be moved at all. Furthermore, the wheels and tyres also carry the full load of the vehicle and give smooth ride. Besides that the wheels and tyres help in operating the vehicle in control with the help of the steering.

5.1 Requirements of the wheel

1. Wheel must to bear the full load acting on the vehicle.
2. It must to make the vehicle weight easy to move and steer easy.
3. It must to help to move the vehicle without the engine.
4. It must to withstand vibrations of the road.
5. It must to withstand the braking force when the brake is applied.

5.2 Properties of the wheel

1. It must be able to withstand the full load of the vehicle.
2. The wheels must be at equal in position while driving the vehicle and at rest as well.
3. The weight of the wheels should be less.
4. The wheels must be easy to remove from the axle and fix in the axle.
5. The wheels must be easy to control by steering mechanism.

5.3 Types of wheels

Generally there are three types of wheel. They are

1. Disc wheel
2. Wired wheel
3. Cast wheel

5.3.1 Disc wheel

This type of wheel is commonly used in all automobiles. The Figure 5.3.1 shows the photographic view of disc wheels. Because its structure is very simple and the construction is very strong and cost effective.



Figure 5.3.1 Shows the Photographic view of Disc Wheels

The disc wheel is made of two parts. They are

1. Rim
2. Disc

The structure of the disc wheel can be seen in Figure 5.3.1. The top part of the rim has the provision for fixing the tyre correctly. The disc is connected with the rim by welding. Depending on the size of the wheel and the disc the disc can be made as an integral part of the rim or separate ones. The part of the tyre which is used for place it on the rim is called as the tyre bead. The tyre cannot be removed or fitted if there is no pit in the rim. The tyre fitting area in the rim is generally in tilting angle between 5° and

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